

2026 논문스터디 SMARLLCLE

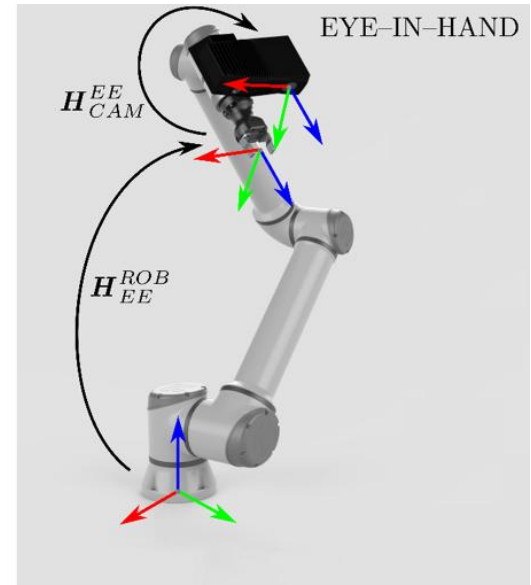
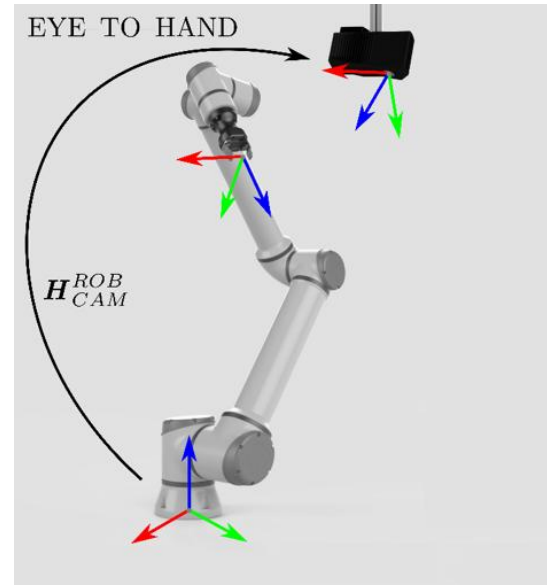
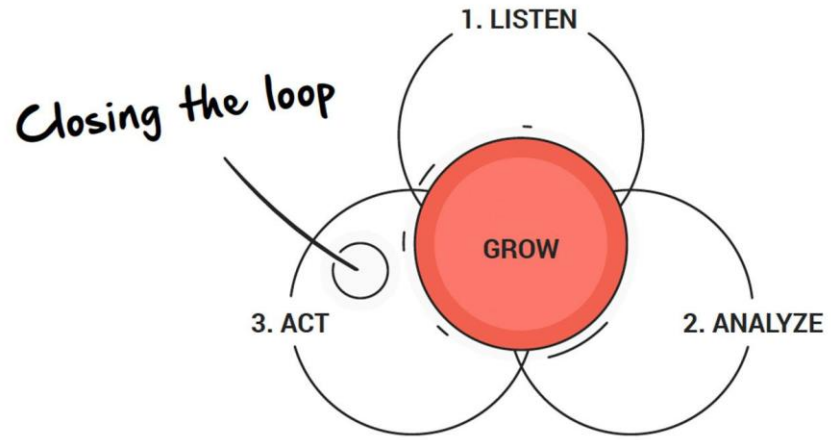
- Learning Fine-Grained Bimanual Manipulation with Low-Cost Hardware

김기현



SMARCLE

Introduction

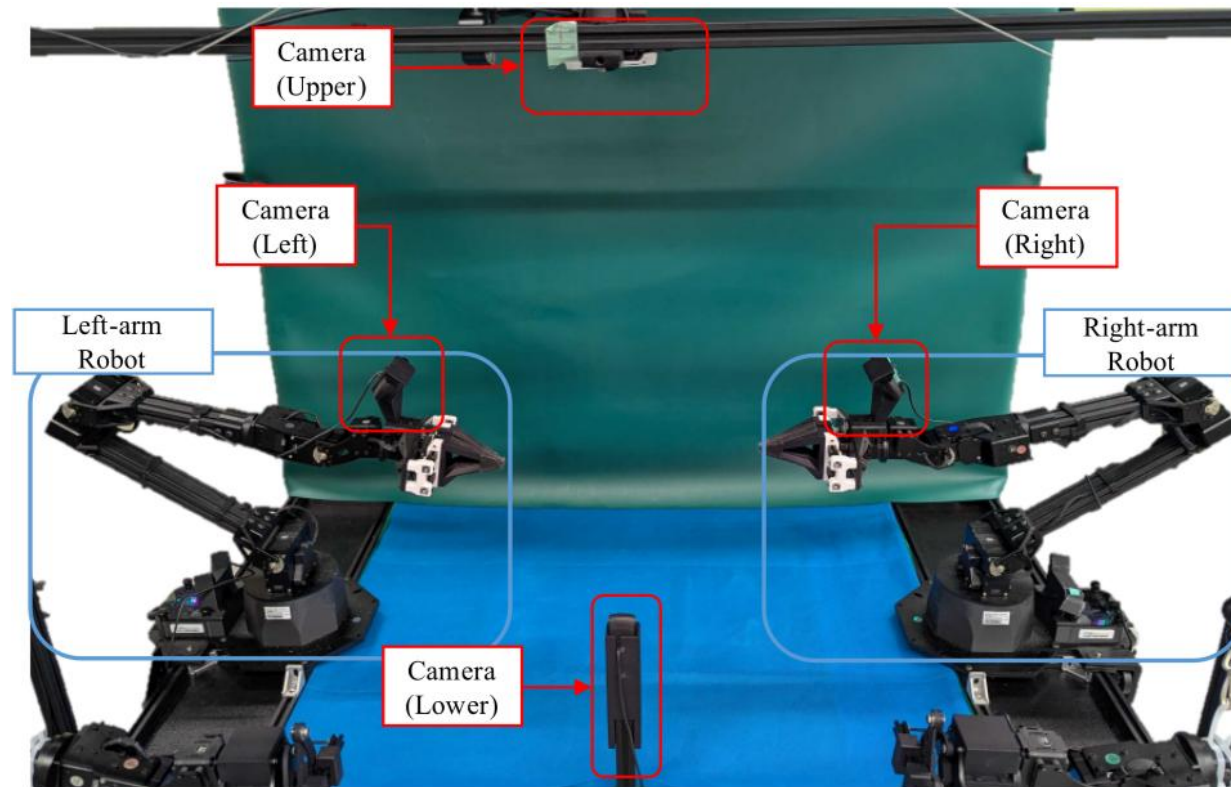


정밀한 조작

- Closed-loop feedback
- 높은 수준의 hand-eye coordination



Introduction



Low-cost Hardware

- Webcam RGB 기반의 end-to-end vision → Action
- 저렴한 로봇팔



Introduction

Learning Fine-Grained Bimanual Manipulation with Low-Cost Hardware

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¹ Stanford University ² UC Berkeley ³ Meta

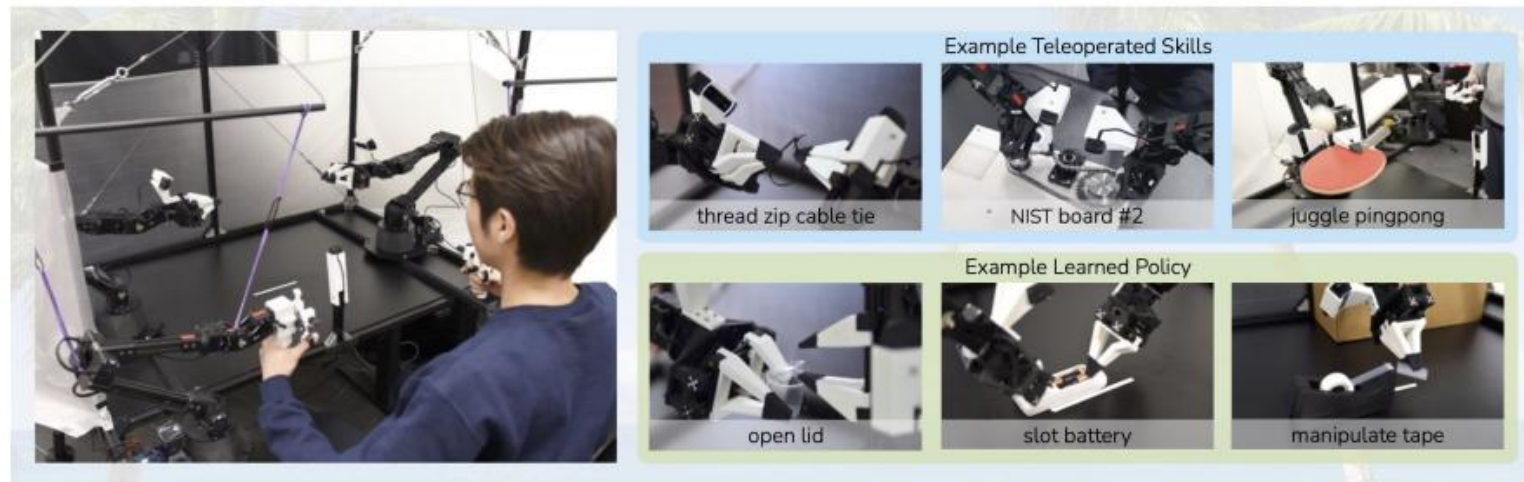
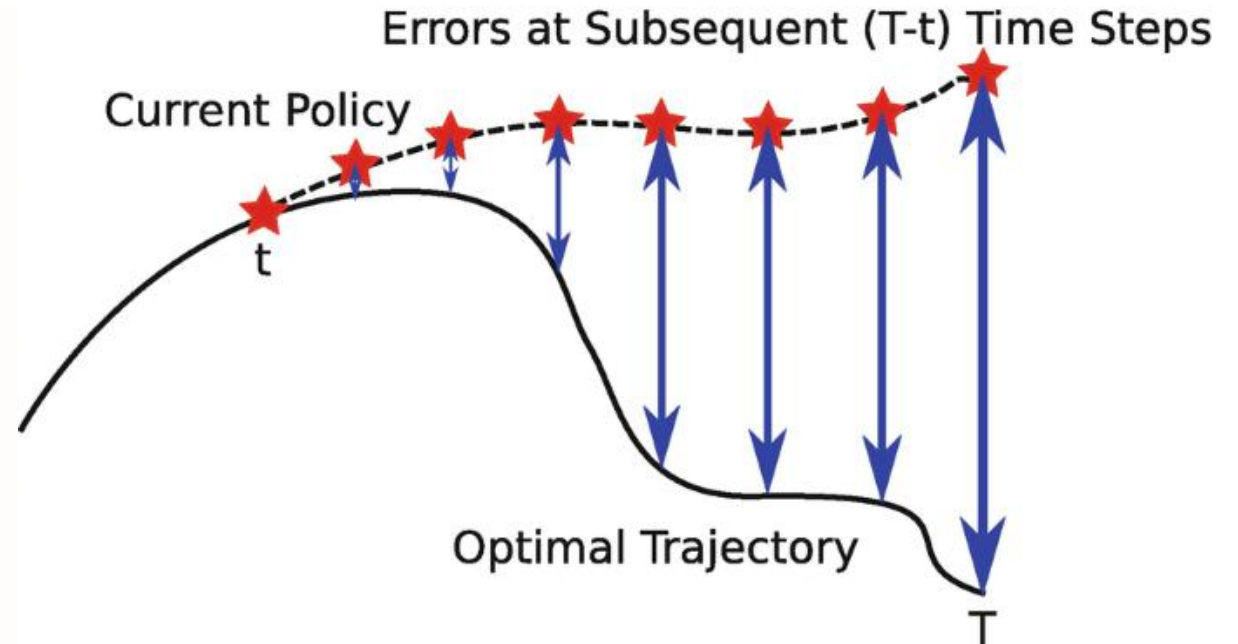
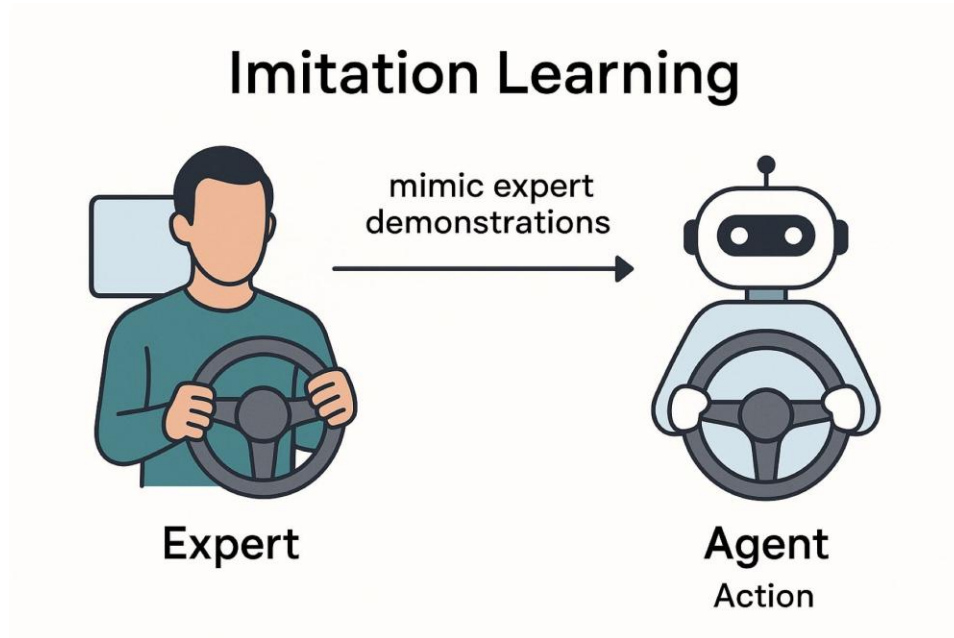


Fig. 1: ALOHA 🐘 : A Low-cost Open-source Hardware System for Bimanual Teleoperation. The whole system costs <\$20k with off-the-shelf robots and 3D printed components. *Left*: The user teleoperates by backdriving the leader robots, with the follower robots mirroring the motion. *Right*: ALOHA is capable of precise, contact-rich, and dynamic tasks. We show examples of both teleoperated and learned skills.

“결과적으로 Teleoperation System & Imitation Learning Alg 제안”



Previous Work

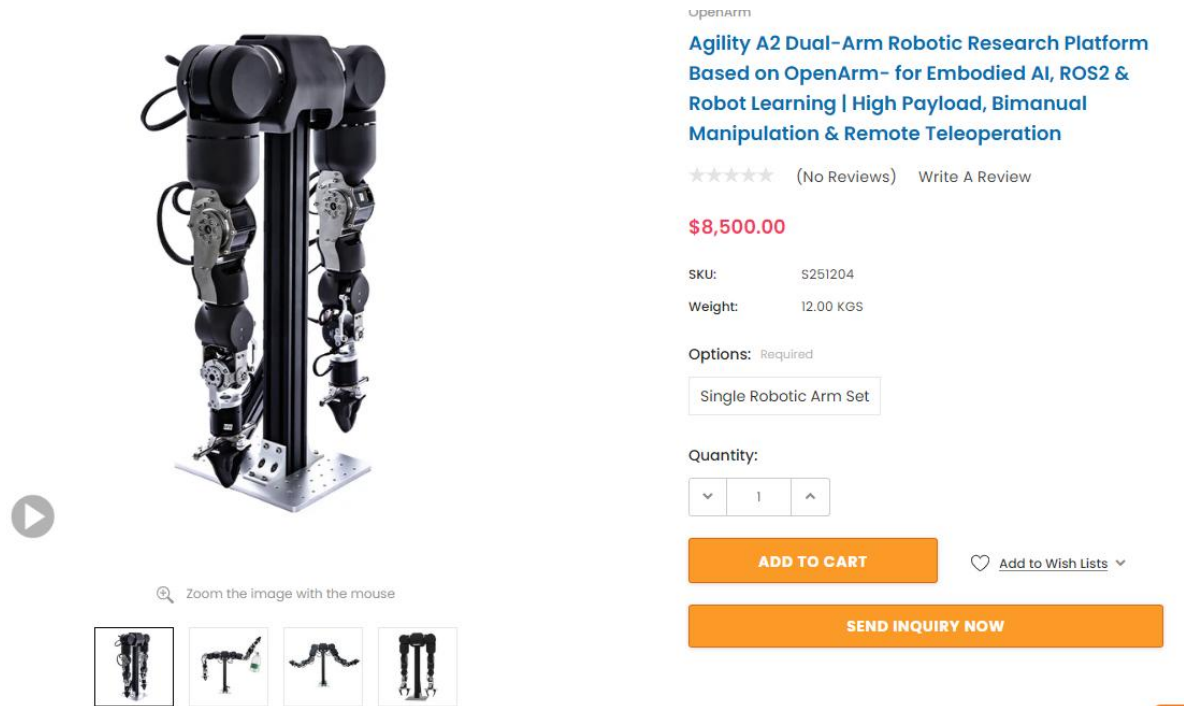


Imitation Learning

- 모방학습이라고 불리며 전문가의 행동을 따라서 학습을 진행
- 누적 오차의 가능성이 야기되어옴.



Previous Work

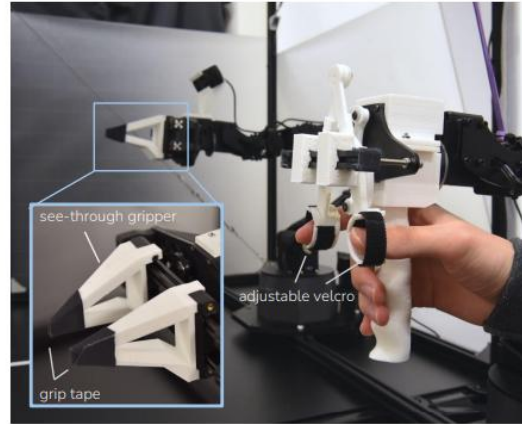
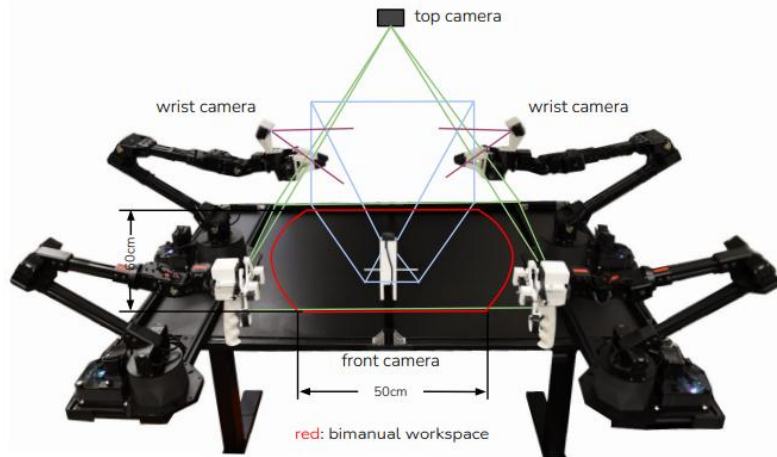


Bimanual Manipulation

- 양팔 로봇은 한팔 로봇보다 가능한 Task가 많음
- 기존에는 양팔 로봇 조작을 위해 비싼 시스템이 사용되었음



ALOHA



ViperX 6dof Arm (follower)

#Dofs	6+gripper
Reach	750mm
Span	1500mm
Repeatability	1mm
Accuracy	5-8mm
Working Payload	750g

Fig. 3: *Left*: Camera viewpoints of the front, top, and two wrist cameras, together with an illustration of the bimanual workspace of ALOHA. *Middle*: Detailed view of the “handle and scissor” mechanism and custom grippers. *Right*: Technical spec of the ViperX 6dof robot [1].

\$20,000 내외의 비용으로 제작 가능

1. Low-cost(저비용)
2. Versatile(다재다능한)
3. User-friendly(사용자 친화적)
4. Repairable(수리가능한)
5. Easy-to-build(쉽게 조립 가능한)



ALOHA



Action Chunking with Transformer

Algorithm 1 ACT Training

- 1: Given: Demo dataset $\mathcal{D}$, chunk size k , weight β .
 - 2: Let a_t , o_t represent action and observation at timestep t , $\bar{o}_t$ represent o_t without image observations.
 - 3: Initialize encoder $q_\phi(z|a_{t:t+k}, \bar{o}_t)$
 - 4: Initialize decoder $\pi_\theta(\hat{a}_{t:t+k}|o_t, z)$
 - 5: **for** iteration $n = 1, 2, \dots$ **do**
 - 6: Sample o_t , $a_{t:t+k}$ from $\mathcal{D}$
 - 7: Sample z from $q_\phi(z|a_{t:t+k}, \bar{o}_t)$
 - 8: Predict $\hat{a}_{t:t+k}$ from $\pi_\theta(\hat{a}_{t:t+k}|o_t, z)$
 - 9: $\mathcal{L}_{reconst} = \text{MSE}(\hat{a}_{t:t+k}, a_{t:t+k})$
 - 10: $\mathcal{L}_{reg} = D_{KL}(q_\phi(z|a_{t:t+k}, \bar{o}_t) \parallel \mathcal{N}(0, I))$
 - 11: Update θ , ϕ with ADAM and $\mathcal{L} = \mathcal{L}_{reconst} + \beta\mathcal{L}_{reg}$
-

Algorithm 2 ACT Inference

- 1: Given: trained π_θ , episode length T , weight m .
 - 2: Initialize FIFO buffers $\mathcal{B}[0 : T]$, where $\mathcal{B}[t]$ stores actions predicted *for* timestep t .
 - 3: **for** timestep $t = 1, 2, \dots, T$ **do**
 - 4: Predict $\hat{a}_{t:t+k}$ with $\pi_\theta(\hat{a}_{t:t+k}|o_t, z)$ where $z = 0$
 - 5: Add $\hat{a}_{t:t+k}$ to buffers $\mathcal{B}[t : t+k]$ respectively
 - 6: Obtain current step actions $A_t = \mathcal{B}[t]$
 - 7: Apply $a_t = \sum_i w_i A_t[i] / \sum_i w_i$, with $w_i = \exp(-m * i)$
-



Chunking

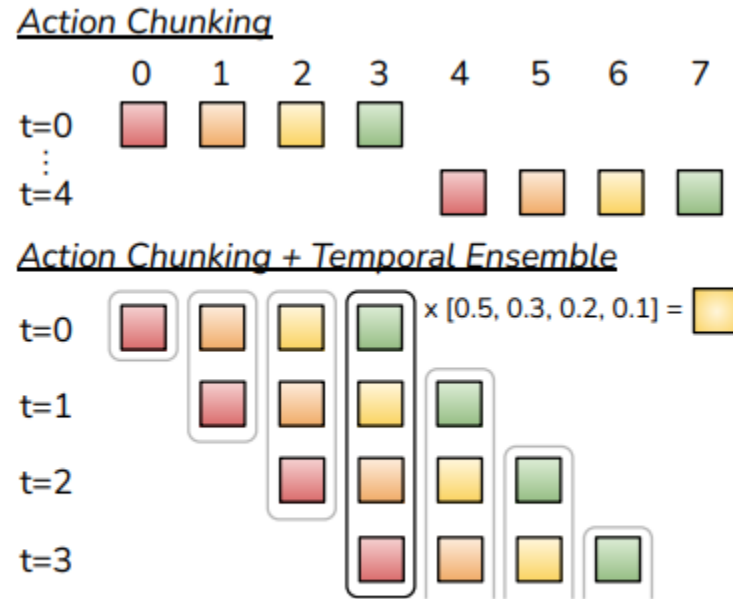


Fig. 5: We employ both Action Chunking and Temporal Ensembling when applying actions, instead of interleaving observing and executing.

- 매 단계마다 k개의 action을 받아냄
- 지수가중결합으로 순간에 대해서 예측된 action을 평균내서 동작



Modeling human data

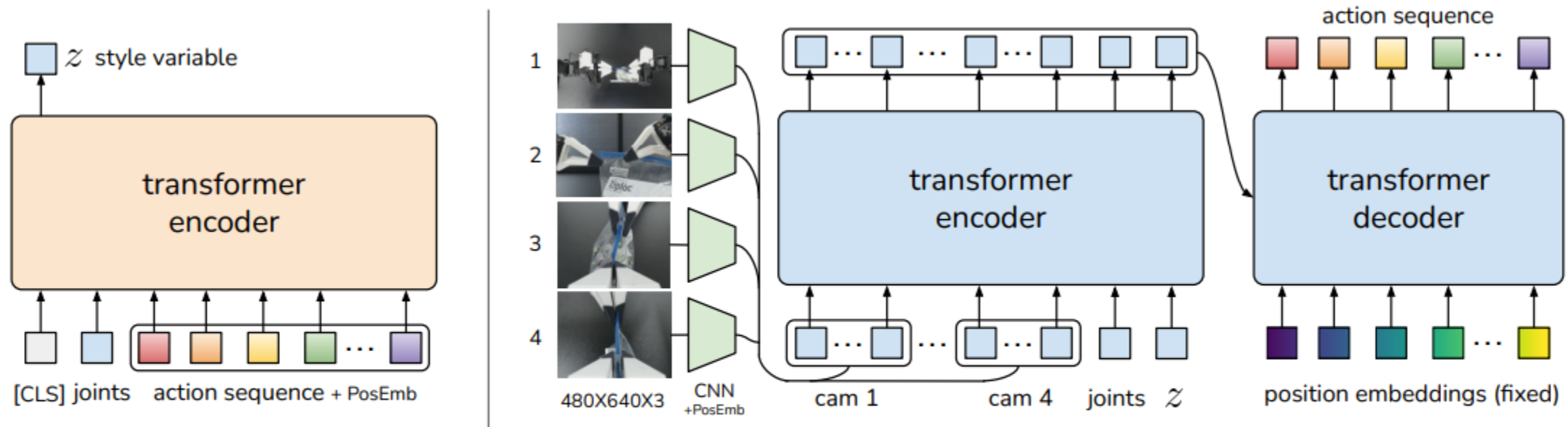


Fig. 4: *Architecture of Action Chunking with Transformers (ACT)*. We train ACT as a Conditional VAE (CVAE), which has an encoder and a decoder. *Left*: The encoder of the CVAE compresses action sequence and joint observation into z , the style variable. The encoder is discarded at test time. *Right*: The decoder or policy of ACT synthesizes images from multiple viewpoints, joint positions, and z with a transformer encoder, and predicts a sequence of actions with a transformer decoder. z is simply set to the mean of the prior (i.e. zero) at test time.

- 잠재 변수를 활용해서 인간 시연의 노이즈를 완화
- Training 단계에서는 z 의 평균과 분산을 예측(학습 효율을 위해 이미지 관찰은 제외)
- Inference 단계에서는 z 를 0으로 설정하고 정책을 결정론적으로 디코딩할 수 있도록



Action Chunking with Transformer

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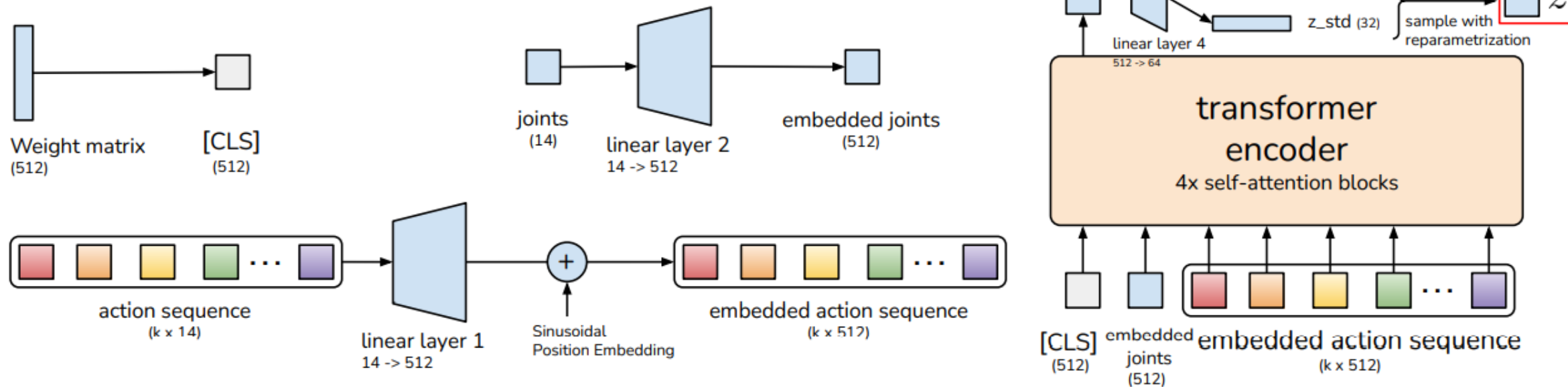
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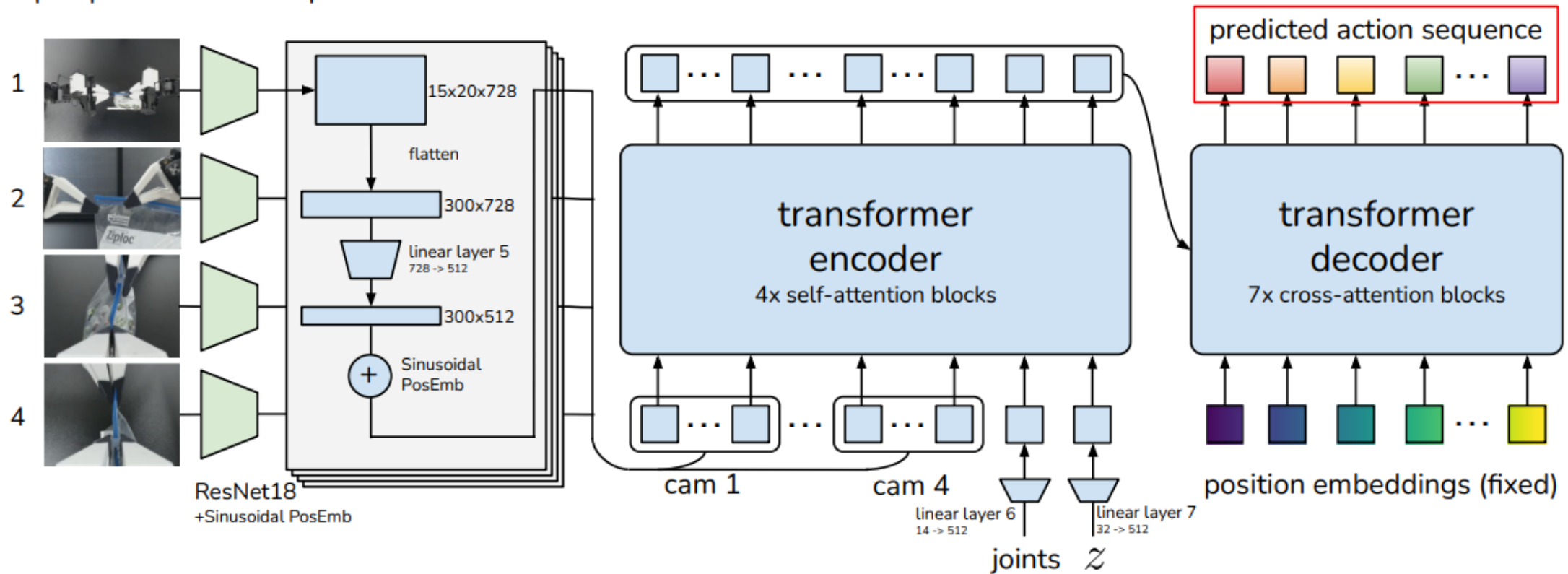
Implementing ACT

Step 2: infer z



Implementing ACT

Step 3: predict action sequence



Experiments

1. 지퍼를 정확하게 잡아서 지퍼백 열기
2. 리모컨 잡고 배터리 섬세하게 밀어 넣기
3. 작은 소스컵의 뚜껑 열기
4. 벨크로 타이를 끼워서 붙이기?
5. 작은 테이프 조각을 판지 상자의 모서리에 붙이는 것
6. 마네킹 발에 신발을 신기고 신발의 벨크로 스트랩으로 고정
7. 큐브 옮기기(sim)
8. 큐브 끼우기(sim)

Real-World Task Definitions



Slide Ziploc: Open the ziploc bag that is standing upright on the table. The bag is randomized along the 15cm white line. It is dropped from ~5cm above the table to randomize the deformation, which affects the height and appearance of the bag. The left arm first grasps the bag body (*Subtask#1 Grasp*) followed by the right arm pinching the slider (*Subtask#2 Pinch*). Then the right arm moves right to unzip the bag (*Subtask#3 Open*).



Slot Battery: Insert the battery into the remote controller. The controller is randomized along the 15cm white line. The battery is initialized in roughly the same position with different rotations. The right arm first grasps the battery (*Subtask#1 Grasp*) then places it into the slot (*Subtask#2 Place*). The left arm presses onto the remote to prevent it from sliding, while the right arm pushes in the battery (*Subtask#3 Insert*).



Open Cup: Pick up and open the lid of a translucent condiment cup. The cup is randomized along the 15cm white line. Both arms approach the cup, and the right gripper gently tips over the cup (*Subtask#1 Tip Over*) and pushes it into the gripper of the left arm. The left arm then gently closes its gripper and lifts the cup off the table (*Subtask#2 Grasp*). Next, the right gripper approaches the cup lid from below and prys open the lid.



Thread Velcro: Pick up the velcro cable tie and insert one end into the small loop on the other end. The velcro tie is randomized along the 15cm white line. The left arm first picks up the velcro tie by pinching near the plastic loop (*Subtask#1 Lift*). The right arm grasps the tail of the velcro tie mid-air (*Subtask#2 Grasp*). Next, both arms coordinate to deform the velcro tie and insert one end of it into the plastic loop on the other end.



Prep Tape: Hang a short segment of tape on the edge of the box. The tape dispenser is randomized along the 15cm white line. First, the right gripper grasps the tape from the side (*Subtask#1 Grasp*). It then lifts the tape and pulls to unroll it, followed by cutting it with the dispenser blade (*Subtask#2 Cut*). Next, the right gripper hands the tape segment to the left gripper in mid-air (*Subtask#3 Handover*), and both arms move toward the corner of the stationary cardboard box. The left arm then lays the tape segment flat on the surface of the box while the right gripper pushes down on the tape to prevent slipping. The left arm then opens its gripper to release the tape (*Subtask#4 Hang*).



Put On Shoe: Put a velcro-strap shoe on a fixed mannequin foot. The shoe pose is randomized along the 15cm white line. First, both left and right grippers pick up the shoe (*Subtask#1 Lift*). Then both arms coordinate to put it on, with the heel touching the heel counter (*Subtask#2 Insert*). Next, the left arm moves to support the shoe (*Subtask#3 Support*), followed by the right arm securing the velcro strap (*Subtask#4 Secure*).

Fig. 6: Real-World Task Definitions. For each of the 6 real-world tasks, we illustrate the initializations and the subtasks.



Experiments

	Cube Transfer (sim)			Bimanual Insertion (sim)			Slide Ziploc (real)			Slot Battery (real)		
	Touched	Lifted	Transfer	Grasp	Contact	Insert	Grasp	Pinch	Open	Grasp	Place	Insert
BC-ConvMLP	34 3	17 1	1 0	5 0	1 0	1 0	0	0	0	0	0	0
BeT	60 16	51 13	27 1	21 0	4 0	3 0	8	0	0	4	0	0
RT-1	44 4	33 2	2 0	2 0	0 0	1 0	4	0	0	4	0	0
VINN	13 17	9 11	3 0	6 0	1 0	1 0	28	0	0	20	0	0
ACT (Ours)	97 82	90 60	86 50	93 76	90 66	32 20	92	96	88	100	100	96

TABLE I: Success rate (%) for 2 simulated and 2 real-world tasks, comparing our method with 4 baselines. For the two simulated tasks, we report [training with scripted data | training with human data], with 3 seeds and 50 policy evaluations each. For the real-world tasks, we report training with human data, with 1 seed and 25 evaluations. Overall, ACT significantly outperforms previous methods.

	Open Cup (real)			Thread Velcro (real)			Prep Tape (real)				Put On Shoe (real)			
	Tip Over	Grasp	Open Lid	Lift	Grasp	Insert	Grasp	Cut	Handover	Hang	Lift	Insert	Support	Secure
BeT	12	0	0	24	0	0	8	0	0	0	12	0	0	0
ACT (Ours)	100	96	84	92	40	20	96	92	72	64	100	92	92	92

TABLE II: Success rate (%) for the remaining 3 real-world tasks. We only compare with the best performing baseline BeT.



Ablations

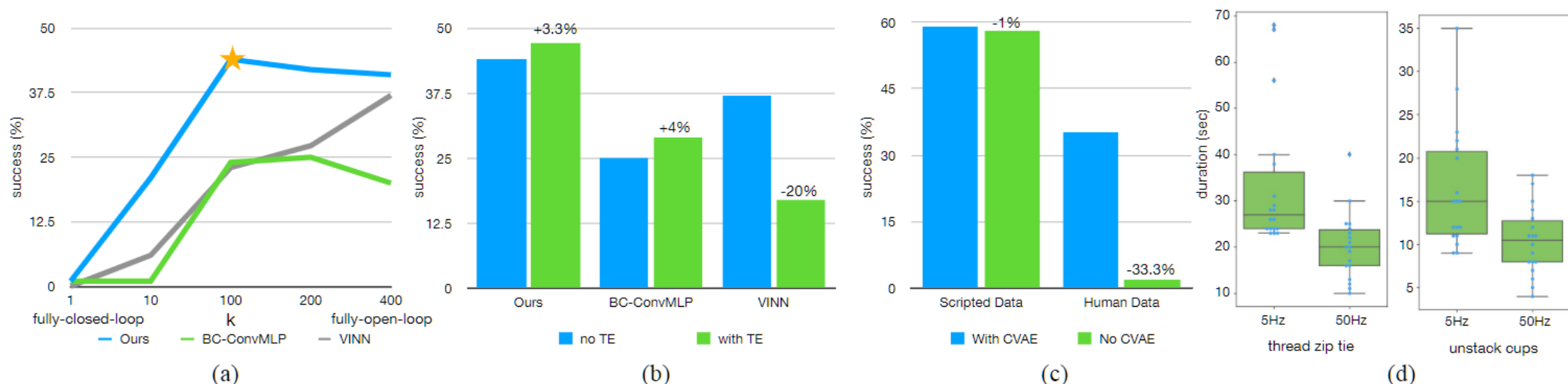


Fig. 8: (a) We augment two baselines with action chunking, with different values of chunk size k on the x-axis, and success rate on the y-axis. Both methods significantly benefit from action chunking, suggesting that it is a generally useful technique. (b) Temporal Ensemble (TE) improves our method and *BC-ConvMLP*, while hurting *VINN*. (c) We compare with and without the CVAE training, showing that it is crucial when learning from human data. (d) We plot the distribution of task completion time in our user study, where we task participants to perform two tasks, at 5Hz or 50Hz teleoperation frequency. Lowering the frequency results in a 62% slowdown in completion time.

- Action chunking & temporal ensembling: non-Markovian적 데모를 더 잘 처리하는 능력
- CVAE: noisy한 인간 데모를 잘 모델링하는 능력
- high-frequency: 주파수에 따른 진행 능력



Limitation and Conclusion

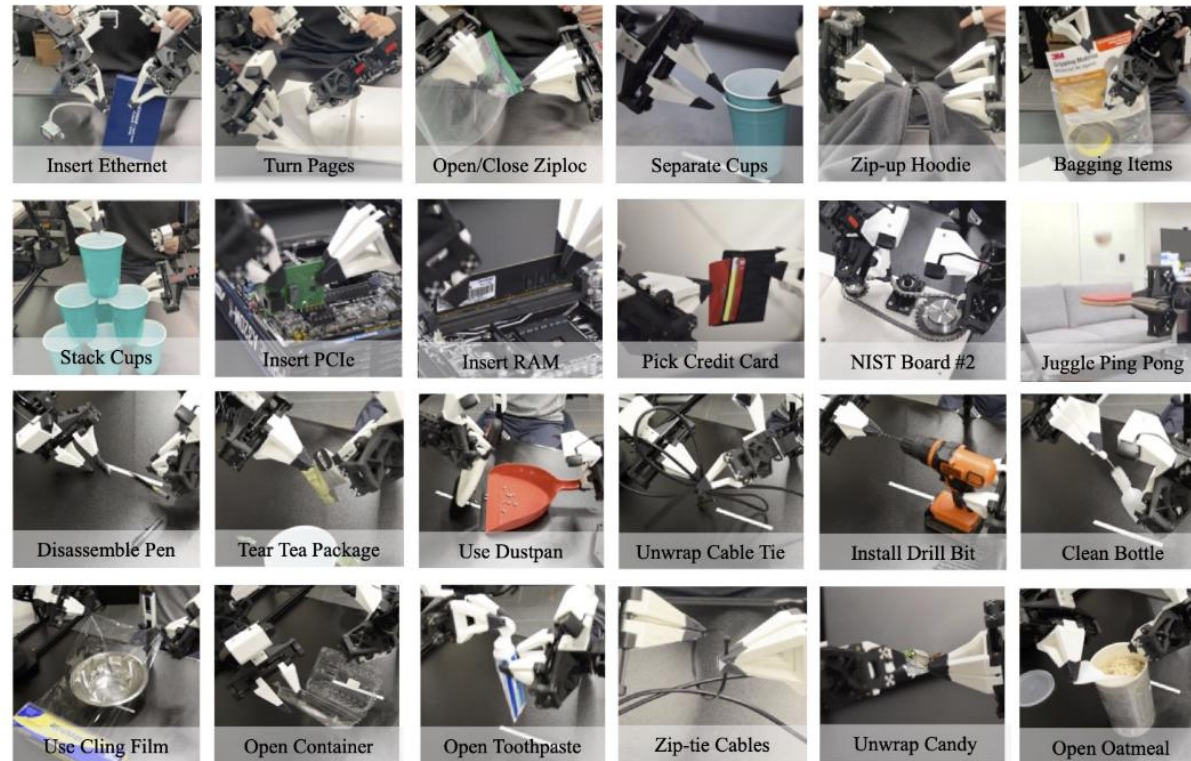


Fig. 9: Teleoperation task examples with ALOHA. We include videos on the [project website](#).

- 저비용 오픈소스 시스템의 중요성을 강조
- 하드웨어와 소프트웨어의 통합 시스템을 선보임





Thank You

S M A R C L E